0.1 Overview

The NA62 experiment is a fixed target experiment located in the north area of CERN with a high energy particle beam provided by the Super Proton Synchrotron. The main goal of the experiment is to measure the branching fraction of the ultra-rare charged kaon decay $K^+ \rightarrow \pi^+\nu\bar{\nu}$ which is of the order of 10^-10 to a 15% precision. Alongside there is a expansive physics program of precision and rare decay measurements as well as searches for forbidden processes and exotic particles.

The experiment employs a decay-in-flight technique to obtain a large number of charged kaon decays, in the order of 10^{13} within the short time frame of a few years. The experimental setup was designed to fulfill all the requirements for the challenging meaurement of $K^+ \rightarrow \pi^+\nu\bar{\nu}$, which involves the detection of the K^+ upstream and the detection of the product π^+, with a missing energy-momentum carried away by the undetected neutrino-antinutrino pair. The measurement of the missing mass is used for kinematic rejection of the main background K^+ decays; $K^+ \rightarrow \pi^+\pi^0$, $K^+ \rightarrow \pi^+\pi^- + \pi^0$ and $K^+ \rightarrow \mu^+\nu_\mu$ by selecting two regions shown in figure 1. The missing mass calculation requires precise timing and position measurement of the K^+ and π^+ candidates, provided by beam (section 0.2.3) and secondary (section 0.2.4) particle spectrometers. The measured 3-momenta of the candidates allows for the reconstruction of the 4-momenta of the Kaon P_{K^+} and the pion P_{π^+} which is used in equation 0.1

$$m^2_{miss} = (P_{K^+} - P_{\pi^+})^2 \tag{0.1}$$

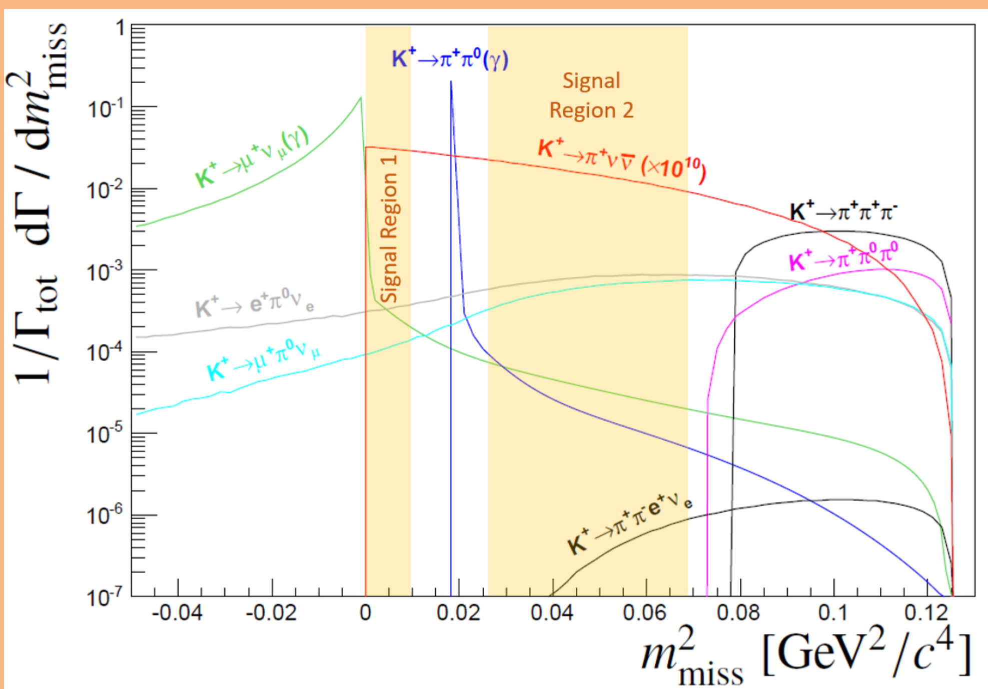

Figure 1: Expected squared missing mass distributions of main kaon decays and $K^+ \rightarrow \pi^+\nu\bar{\nu}$ (scaled by 10^{10} for for visability). The solid coloured areas show the signal regions used to kinematically reject the domiant kaon decays.

The rest of this chapter will cover how NA62 fulfills the abovementioned requirements. Section 0.2.1 will describe the NA62 beam-line, upstream of the NA62 detector. The NA62 detector comprises a number of sub-detectors, which will be described in Section **??**, with the trigger and data acquisition system (TDAQ) following in Section 0.3

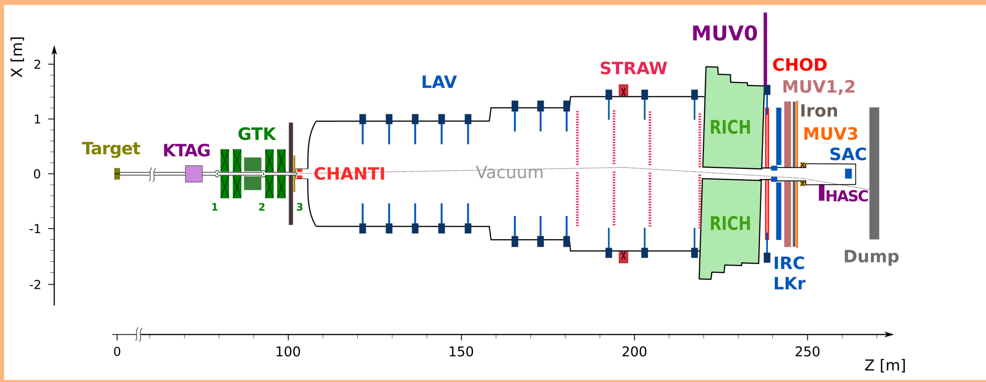

Figure 2: X-Z diagram of NA62 detector setup for Run 1

0.2 The NA62 Detector

0.2.1 Beam-Line

The experiment at CERN uses a hadron beam extracted from the SPS with a nominal momentum of 400 GeV/c, directed onto a 400 mm long, 2 mm diameter berylium rod that forms the fixed target (T10) located at 0 m in figure 3. The beam derived from the T10 target, referred to as K12, is a high-intensity unseparated hadron beam with a nominal momentum of 75 GeV/c. The beam momenta was chosen to maximise the fraction of kaons with respect to the other beam components at the entrance to the detector.

The elements of the K12 beam-line are shown in Figure 3 and comprise a number of components. First are three quadrupole magnets (Q1, Q2 and Q3), used to focus the beam with a large solid angle acceptance of $\pm$2.7 mrad horizontally and $\pm$1.5 mrad vertically at a 75 GeV/c central momentum. Next follows an achromat (A1), which selects a beam of a 75 GeV/c momentum with a 1 % rms momentum. The A1 achromat comprises four dipole magnets that vertically deflect the beam, the first two which deflect the beam 110 mm downwards, while the final two deflect the beam upwards towards the original axis. Between these pairs of magnets, there are a pair of water-cooled beam-dump units (TAX1 and TAX2) with a set of holes to select the required 75 GeV/c momentum beam, while absorbing unwanted secondary beam particles and any remaining primary protons. TAX1&2 can be closed to act as a secondary target for use in dump mode or for safety when access is required downstream. At the focus between the TAX1&2, a radiator consisting of tungston plates with a thickness of up to 5 mm is introduced into the path of beam. The radiator is optmised to reduce positron energy while minimising the loss of energy to hadrons. A triplet of quadrupoles (Q4, Q5, Q6) follows, which refocuses the beam in the vertical dimension and produces a parallel beam in the horizontal dimension. The space between the quadrupoles are occupied by a pair of collimators (C1, C2), which redefines the horizontal and vertical acceptance for the beam. The collimator after the Q6 quadrupole (C3) is employed to absorb positrons that had their momenta degraded by the radiator and again redefines the acceptance in the vertical dimension.

The beam next passes through a 40 mm field-free hole in a set of iron plates, placed between three 2 m dipole magnets (B3). The vertical magnetic field in the iron plates is used to deflect muons of both signs away from the beam. Any deviation in the beam due to stray fields in the beam hole are canceled out by a pair of steering dipoles (TRIM2 and TRIM3). Two quadrupoles (Q7, Q8) follow and ensure that the beam is parallel before the KTAG which employs a differential Cherenkov counter (CEDAR), described in Section 0.2.2, which requires the beam to be parallel. Before the KTAG are a pair of collimators (C4 and C5) used to absorb particles in the tails of the beam.

For beam monitoring two pairs of filament scintillator counters (FISC 1, 3 and FISC 2, 4) are installed at either end of the KTAG. Used in coincidence the beam divergence can be measured and used to tune the divergence to zero.

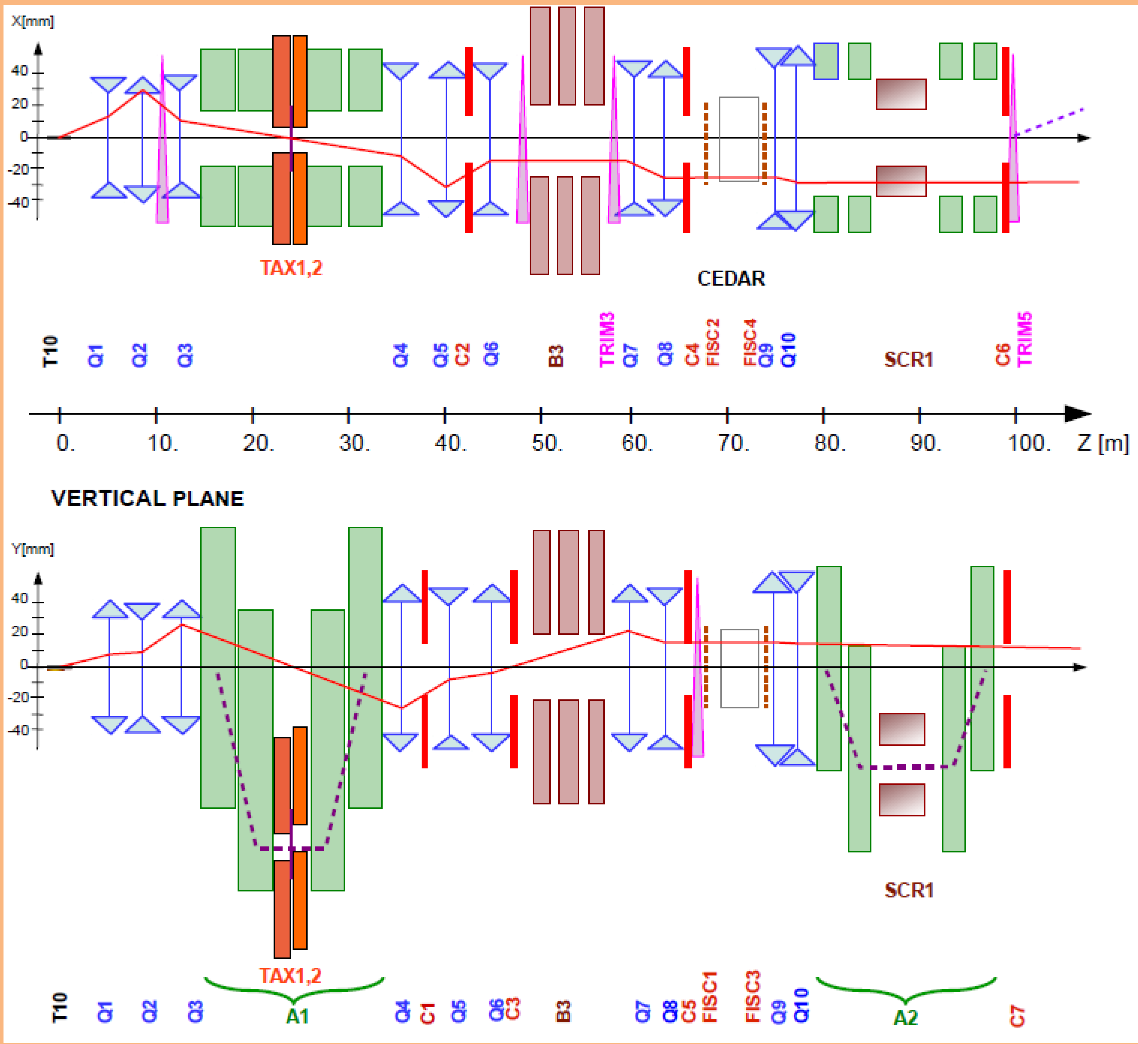

Figure 3: Schematic of K12 beamline from T10 target at 0m to the entence of the NA62 decay region at 102.4m(2).

0.2.2 KTAG

0.2.3 GigaTracker (GTK)

0.2.4 Straw Spectrometer (Straw)

0.2.5 RICH

0.2.6 Photon Detection System

0.3 Triger and Data Acquisition System (TDAQ)

0.4 Things to be moved

From the T10 target, the charged kaon beam travels through the beam pipe to the experimental hall ECN3, where the first sub-detector is found. A differential Cherenkov detector tags the charged kaons in the beam. Cherenkov light is produced by charged particles passing through a CERN CEDAR vessel filled with Nitrogen gas at a pressure of around 1.75 bar (1). Due to the high intensity and tight time resolution requirements employed at NA62, the original photo-detectors (PMT's) and front-end electronics were inadequate. The KTAG has been developed for the NA62 to meet the high kaon rate requirements and time resolution requirements of $\mathcal{O}(100ps)$.
Downstream of the KTAG is the beam spectrometer named the Giga Tracker (GTK). The GTK

comprises 4 silicon trackers stations to measure the beam's position, time and momentum. As the GTK is directly in the beamline, vetoing any particles produced from inelastic interactions is essential. This is done by the Charged Anti-coincidence detector (CHANTI) located slightly downstream of the final GTK station. The CHANTI detects charged particles using polystyrene base scintillators.
The main decay volume is located downstream of the CHANTI and is in a large vacuum tube shown in figure 2. This is where kaons are allowed to decay if they are to be within the acceptance of the various sub-detectors. Downstream from this is the STRAW Spectrometer which provides kinematic measurements of the charged products of the kaon decays. It consists of four spectrometer chambers and a dipole magnet (MNP33) with two of the chambers located before the magnet and two after. A more detailed description of this detector is provided in section 0.4.
The Ring Imaging Cherenkov (RICH) follows the last straw chamber and is a 17 m long neon-filled vessel. It is designed to distinguish between pions and muons and is part of the Particle Identification system (PID). This is also complemented by the Muon Veto (MUV) system comprising two hadronic calorimeters (MUV1 and MUV2)
A Photon-Veto system provides coverage of photons produced from the decay region up to 50 mrad with respect to the z-axis. This includes:

- 12 Large Angle Vetos (LAVs) placed along the decay region, between STRAW spectrometer chambers and after the RICH. Providing coverage of photons emitted at angles from 8.5 to 50 mrad (2).
- Liquid Krypton calorimeter (LKr) which is located after the RICH and was used in the predecessor experiment NA48.
- Small Angle Veto (SAV) system, comprising of two shashlyk calorimeters named the Small Angle Calorimeter (SAC) and the Intermediate-Ring Calorimeter (IRC) which provides photon veto coverage for small angles.

There are several other detectors, such as the MUV3, which provides fast signals for use in the L0 trigger system and a pair of Charged Particle Hodoscopes (CHODs), which also provide input to the L0 trigger system. Some additional veto detectors, such as MUV0 and the HASC are used to for analysis.
For the new run (after LS2), there have been several additions to the detector setup. The ANTI-0 detector is a hodoscope placed downstream of the CHANTI before the decay volume, used to veto halo particles (mostly muons) that enter the decay volume (3). A second detector system has been implemented between the GTK stations to veto any decay products of kaons decaying between the GTK stations, named the Veto Counter (VC), it is made up of three stations of scintillators between GTK stations 3 and 2.
Two sub-detectors, KTAG and STRAW spectrometer, will be described in more detail due to their relevance to the studies that are planned for my PhD. The KTAG-CEDAR has been selected due to my involvement in the CEDAR-H test beam. The STRAW spectrometer has been selected due to my participation in the development of flexibility of the sub-detector and its subsequent use in the symmetric NA62 study later on in this report.

KTAG

As mentioned previously, kaons are identified in the beam by the CEDAR-KTAG subdetector. The charged beam produces Cherenkov light when passing through the Nitrogen gas-filled CEDAR. The light produced is then reflected onto a ring-shaped diaphragm by the Mangin mirror, located at the end of the CEDAR vessel, which is a special geometry to produce the ring shape needed to distinguish particles. Between the mirror and the diaphragm, the light passes through a chromatic corrector lens that corrects for the dispersion caused by the gas. The diaphragm only allows the light of a certain radius to pass through, allowing the selection of certain particles corresponding light. The particle is selected by altering the pressure of the gas. After the diaphragm the light passes through several quartz windows into the KTAG enclosure.

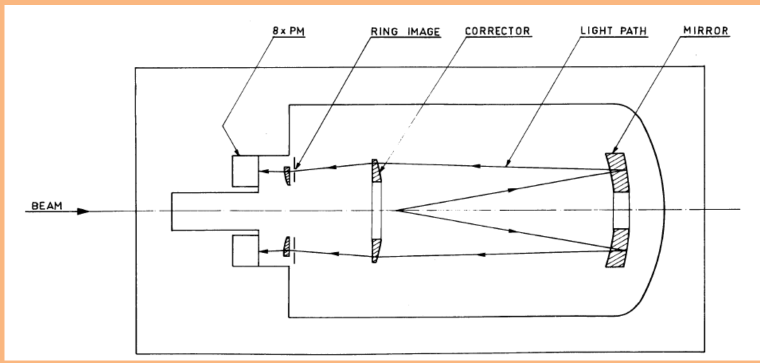

Figure 4: Diagram describing the CEDAR optics (1)

The light is reflected by eight spherical mirrors located around the beam pipe into eight light boxes, which are seen in figure 5. Each lightbox (commonly referred to as sectors) contains 48 photomultipliers converting the light into electrical signals that are then collected by the front-end electronics.
At the start of a data-taking period, a pressure scan is completed to isolate the kaon peak from the other constituents of the beam. Through tests and pressure scans, the optimal pressure for kaons is set at 1.75 bar. Considering the quartz windows and the Nitrogen gas, the total material path of the beam is $3.5 \times 10^{-2} X_0$ (2).

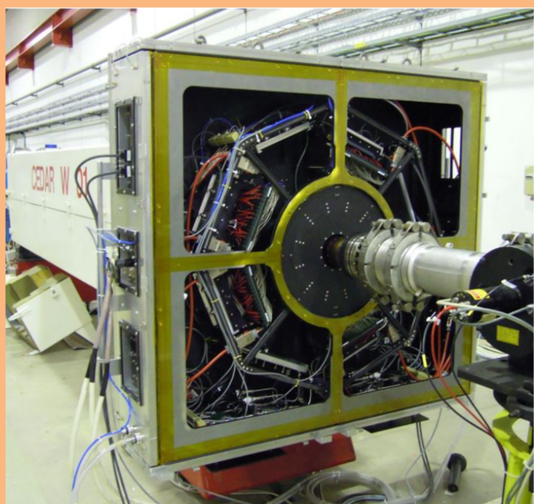

Figure 5: Photo of CEDAR and KTAG installed in the NA62 beamline

Improving the material budget is possible by replacing Nitrogen with Hydrogen at a higher pressure of 3.9 bar, resulting in a reduced material path of $7 \times 10^{-3} X_0$.

STRAW Spectrometer

The STRAW Spectrometer is used for kinematic measurements of charged decay products and comprises of four separate chambers and one dipole magnet (MNP33). The chambers are located downstream from the decay region, with chambers 1 and 2 separated by 10 m with the MNP33 magnet placed downstream of chamber 2. Chambers 3 and 4 are located downstream of the MNP33 with a separation of 14 m. Each chamber comprises four views, containing four layers of straws in a pattern shown in figure 6. The straws are 2.1 m long drift tubes that are about 9.75 mm in diameter containing an $ArCO_2$ gas mixture and are similar to proportional counters. As a particle passes through the gas, it ionises the gas producing a signal in the central wire. This allows for a measurement of the spatial position of the particle, which, along with the MNP33 allows for momentum calculations of the particles.

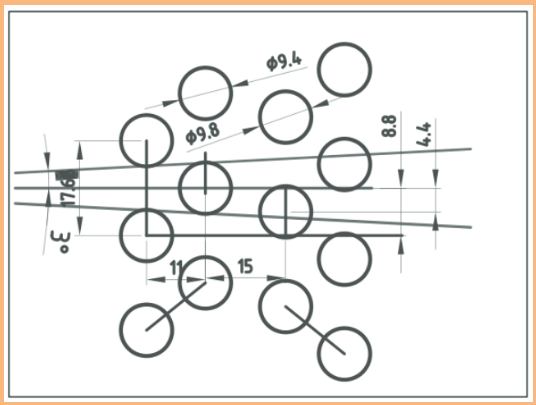

Figure 6: Diagram of straw placement in each view (4)

Bibliography

[1] C. Bovet, R. Maleyran, L. Piemontese, A. Placci, and M. Placidi, "The Cedar Counters for Particle Identification in the SPS Secondary Beams: A Description and an Operation Manual," 12 1982.

[2] NA62Collaboration, "The beam and detector of the NA62 experiment at CERN," *Journal of Instrumentation*, vol. 12, no. 05, pp. P05 025–P05 025, may 2017. [Online]. Available: https://doi.org/10.1088%2F1748-0221%2F12%2F05%2Fp05025

[3] H. Danielsson *et al.*, "New veto hodoscope ANTI-0 for the NA62 experiment at CERN," *JINST*, vol. 15, p. C07007. 8 p, Jul 2020, iNSTR20: Instrumentation for Colliding Beam Physics. [Online]. Available: https://cds.cern.ch/record/2723429

[4] A. Sergi, "Na62 spectrometer: A low mass straw tracker," *Physics Procedia*, vol. 37, pp. 530–534, 2012, proceedings of the 2nd International Conference on Technology and Instrumentation in Particle Physics (TIPP 2011). [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1875389212017348